Abstract

Most collaboration software treats meeting transcripts as afterthoughts that are generated, summarized, and filed away once the conversation concludes. This proposal introduces **SharedTexterial**, a multi-user immersive environment where spoken conversation is continuously converted into text objects. Participants can manually grab, resize, merge, and arrange these objects while a large language model (LLM) quietly builds a shared, 3D map of how the group's ideas connect. The system extends Microsoft Research's *Texterial* (2026) concept—treating text as material for a single writer—into a collaborative setting. A controlled study is proposed to compare SharedTexterial against conventional VR whiteboards and VR-plus-automatic-transcript baselines to see if handling words as physical material improves how a group thinks together.

1. Motivation and Background

- **The Problem:** In long working meetings, conversations are where thinking happens, but the live, shared record of how ideas connect is often lost. Notes lag behind discussions, whiteboards require manual upkeep, and AI summaries replace group sense-making with a model's interpretation.
- **The Inspiration:** *Texterial* (Shen et al., 2026) allows single writers to sculpt or grow text like clay or plants.
- **The Solution:** SharedTexterial brings this concept to groups, turning spoken words into physical objects that can be manipulated by anyone in the room, while an LLM builds a 3D knowledge structure that requires human approval before editing.

2. Related Work

SharedTexterial sits at the intersection of several established research threads:

- **Collaborative Text Materiality:** Pushes back against linear chat boxes (e.g., Graphologue, Sensecape, Luminate) by translating single-user text materiality into a multi-user space.
- **Group Sensemaking:** Inherits the distinction between a group's product and their process (e.g., CoSense), using text-as-material to address the erosion of common ground in distributed teams.
- **Immersive Analytics:** Applies spatial knowledge structures to live, unstructured conversations rather than pre-prepared datasets.
- **AI-Assisted Meeting Notes:** Addresses the need for multi-level recaps by allowing live, editable semantic trees.
- **Tangible Metaphors:** Builds on Ishii and Ullmer's (1997) foundational arguments that digital information is easier to think with when coupled with physical graspability.

3. Research Questions

1. **RQ1:** Does collaborative manipulation of live text materials improve group idea generation, compared with conventional collaborative note-taking and whiteboarding?
2. **RQ2:** Does a shared, continuously evolving semantic structure improve participants' understanding of how a discussion was organized?

3. **RQ3:** How does LLM-mediated organization of the group's material affect participants' trust in the system, their willingness to participate, and their sense of ownership over the resulting ideas?
4. **RQ4:** Does spatial, embodied interaction with language improve participants' longer-term recall of what a meeting concluded?

4. The Proposed System: SharedTexterial

4.1 From Speech to Material

- Microphones feed a streaming speech recognizer (like Whisper) with speaker diarization.
- An LLM segments the stream into meaningful chunks (clauses/sentences) and spawns them as text objects in 3D space.
- Supported gestures mirror tangible interaction primitives: *grab, pin, cluster, stretch, combine, connect, and delete*.

4.2 The Shared Semantic Tree

- A 3D tree grows centrally, representing root themes, subtopics, evidence, open questions, and action items.
- The tree is zoomable for macro and micro views.
- The LLM proposes structural attachments, but participants must manually accept them.

4.3 Human-in-the-Loop AI Assistance

- The AI performs clustering, summarization, and duplicate/contradiction detection in the background.
- It is restricted to suggesting and highlighting; it cannot silently restructure anything a human has built.

5. Study Design

Participants and Conditions

A between-subjects study with 24–36 participants organized into teams of 3–4, randomly assigned to one of three conditions:

Condition	**Design**
A — VR Whiteboard (baseline)	A conventional immersive whiteboard where participants sketch and place virtual sticky notes freehand, with no automatic transcription or AI structuring.

B — VR + Automatic Transcript	The same VR room with automatic transcription and an AI-generated summary after the discussion, isolating the effect of AI note-taking without physical manipulation.
C — SharedTexterial	The full system: manipulable text objects and a live-editable, LLM-assisted semantic tree.

Tasks and Measures

- **Tasks:** Open-ended planning (e.g., smart-city initiatives, disaster-response protocols).
- **Objective Measures:** Idea count/diversity, completion time, revisions, graph structural complexity, interaction frequency, and distribution of speaking/manipulation time.
- **Subjective Measures:** NASA-TLX, System Usability Scale (SUS), Creativity Support Index, Presence Questionnaire, and custom scales for shared understanding, AI trust, and idea ownership.

6. Technical Architecture

Layer	Components
Rendering & interaction	Unity; Meta XR SDK; Quest 3 hand tracking (no controllers).
Networking	Photon Fusion or Unity Netcode for state synchronization.
Speech capture	Streaming ASR (Whisper) with speaker diarization.
Language understanding	Gemini flash-class model for segmentation, clustering, and detection and minimal latency.
Knowledge storage	Vector database paired with a lightweight knowledge-graph layer.

7. Deliverables and Timeline

The project spans 20 months across six distinct phases:

Phase	Timeframe	Key Deliverables
1. Formative Study & Design	Months 1–3	Interviews and co-design with 8–10 users; interaction-design specification.
2. Prototype I: Single-User	Months 3–7	Working speech-to-text pipeline validated outside VR; Quest 3 single-user build.
3. Prototype II: Multi-User	Months 6–10	Networked synchronization; working shared semantic tree; internal pilot (2-3 triads).
4. Baselines & Pilot Study	Months 10–13	Builds for Conditions A & B; IRB approval; small pilot study ($n \approx 9$).
5. Full Controlled Study	Months 13–17	Completed study ($n = 24–36$); cleaned datasets (logs, videos, transcripts).
6. Analysis & Dissemination	Months 17–20	Analysis report; CHI paper draft; open-source release with video figure.

8. Risks and Future Directions

- **Risks:** System latency could ruin the material metaphor; VR immersion itself acts as a confound; gesture manipulations might not scale past 4–5 users; and the LLM may struggle to parse fast, overlapping speech.
- **Future Directions:** Adapting the tool for recurring meetings to build "institutional memory" over time, and exploring hybrid sessions where some users join via VR while others use flat screens.